

Epistemic Collapse in Multi-Agent Systems

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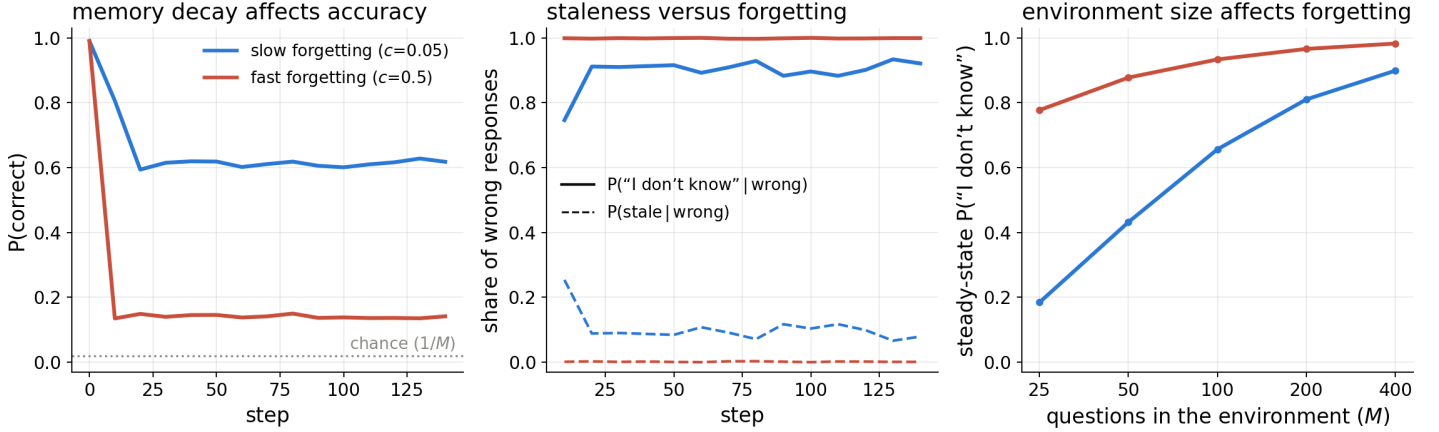


Figure 1: Epistemic collapse. All agents start with knowledge of correct answers to all M questions in the environment. As system evolves, agent knowledge is updated only via direct observations from the environment and retrieved via top- k_{ctx} relevance $\langle q, k \rangle e^{-c \text{ age}}$. *Left.* Temporal component of memory effects an agent’s ability to answer a question correctly. *Center.* The proportion of forgotten answers versus stale answers is higher in fast forgetting agents. *Right.* A large environment causes epistemic collapse in both fast forgetting and slow forgetting agents.

1 Motivation

Information is being produced at unprecedented rates, and increasingly it is produced, relayed, and consumed by generative agents rather than people. An agent embedded in such a flow faces a compounding problem: whatever it has learned is decaying in relevance while new information arrives fast enough to crowd out the old. Communication is the standard remedy — agents ask each other, repeat each other, and increasingly write to shared artifacts — but communication moves evidence around; it does not keep evidence current. Our goal is to understand the *benefits and limits of communication in a high-speed environment*: when does talking to peers, or maintaining a shared record, actually keep a population of agents knowledgeable, and when does it merely keep them *confident*? Building on our earlier work on monitoring multi-agent systems (*Control Charts for Multi-agent Systems*, arXiv:2605.11135), we study these questions in a real network of LLM agents and in a mechanistically faithful simulator, organized around a failure mode we call *epistemic collapse*.

2 Setting

A system of $N = 10$ gpt-4o-mini agents evolves for $T = 500$ steps. Its components, by factor:

Environment. An external store holds the current true answer to each of $M = 50$ questions (drawn from Natural Questions). No agent knows any truth except by consulting the environment. Every question is temporal: question q is revised at rate λ_q per step, with revisions appending a counter to the answer so that stale copies are detectably wrong. Speeds are heterogeneous across questions: λ_q is log-normal with mean $\bar{\lambda}$ and dispersion 1, so a typical world mixes questions an order of magnitude faster and slower than $\bar{\lambda}$.

Agents. Each agent holds an unbounded memory of question-answer pairs, each tagged with its insertion time and provenance (firsthand observation vs. received). Retrieval for question q

returns the top $k_{\text{ctx}} = 3$ entries by relevance $\langle q, k \rangle e^{-c \text{ age}}$ on the insertion clock: old entries are never deleted, they are *crowded out* by newer insertions, so forgetting is activity-dependent — a busy agent forgets faster. The decay rate c is the agent-side lever. Answering is done by the LLM from the retrieved context, under instruction to abstain (“I don’t know,” IDK) without grounding. Under the *verbatim relay* protocol an agent may emit only an answer that originated in the environment, or IDK — no paraphrase, so relayed content does not drift.

Communication. Agents sit on a fixed graph of specified mean degree. A peer ask targets a question the asker currently lacks, drawn uniformly; the peer’s reply (possibly IDK) inserts into the asker’s memory as received evidence. Each agent is seeded with 5 of the 50 questions at $t = 0$.

Observation. Each agent spends exactly $B = 11$ actions per step, split as E environment queries and $K = B - E$ peer asks. An environment query returns the current truth, exact and freshly stamped. Observation is the only action that *creates* current evidence; everything else redistributes it.

Measurement. On a fixed 20-question probe panel, averaged over the final 100 steps, we track four quantities: agent-level $P(\text{IDK})$ and $P(\text{correct})$ (mean over members) and system-level $P(\text{IDK})$ (no member answers) and $P(\text{correct})$ (some member is correct). These nest — system $\text{IDK} \leq \text{agent IDK}$, agent $\text{correct} \leq \text{system correct}$ — so both levels share an axis.

A measured surrogate. Real runs cost money; most of our experiments run on a simulator with *zero fitted parameters*. Under verbatim relay the LLM’s causal role reduces to a gate — given retrieved context, emit the stored payload or abstain — and that gate is measurable: probing gpt-4o-mini across the (finitely many, up to exchangeability and similarity bucketing) context states yields a step function (target present $\Rightarrow$ answer w.p. ≈ 0.95 –1; absent $\Rightarrow$ IDK w.p. ≈ 0.95 –1; parametric leakage ≈ 0). Replacing the LLM call with this measured table

reproduces the real system across all cells of the no-mimesis factor grid and the mimesis grid of Sec. 3 with median $|\Delta| = 0.01$ on all four metrics (Fig. 2).

2.1 Epistemic collapse

We say an agent has *epistemically collapsed* on a question set when it can no longer ground answers and produces only abstentions; the *system* has collapsed when no member can answer. Collapse has a quieter twin: an agent whose evidence has expired but still surfaces it answers *confidently and wrongly* — availability survives, accuracy dies. Both matter, and every mechanism below prices them differently.

Figure 1 shows collapse from the best possible start: every agent warm-started with current, firsthand knowledge of everything. Accuracy falls from ≈ 1 to a plateau set by memory decay (~ 0.61 at $c = 0.05$, ~ 0.14 at $c = 0.5$; left). Conditioned on being wrong, the fast-decay system fails purely by *forgetting* (100% abstention); the slow-decay system’s errors are $\sim 90\%$ forgetting and $\sim 10\%$ *confident staleness* — rising to $\sim 25\%$ in the early transient — with the stale share concentrated on the fastest-revising questions. Abstention is insensitive to how fast any fact revises, but highly sensitive to how many facts there are: steady-state $P(\text{IDK})$ climbs from 0.18 to 0.90 as the environment grows from 25 to 400 questions at fixed budget (right) — breadth, not speed, silences the population, because every new fact’s insertions crowd the others out of retrievable memory. The same regimes appear in the real gpt-4o-mini factor grids (project artifacts): decay ($c : 0.05 \rightarrow 1.0$ takes agent IDK $0.35 \rightarrow 0.89$), observation starvation ($E : 6 \rightarrow 1$ takes system IDK $0.29 \rightarrow 0$), environment speed ($\lambda : 0.001 \rightarrow 0.1$ takes system accuracy $0.93 \rightarrow 0.20$ without moving abstention).

The rest of this note studies the two natural *mitigations*, both forms of communication: (1) **sharing secondhand information** (mimesis) — letting agents repeat answers they received rather than observed; and (2) **books** — a shared artifact agents write to and read from. Both run on the validated platform of Sec. 2.

3 Mitigation 1: sharing secondhand information

The mimesis grid repeats the factor design of Sec. 2 with one change: received answers may ground an answer (still verbatim — agents repeat, they do not paraphrase). Figure 3 overlays the two grids at the agent level.

Mimesis raises the floor. At baseline, agent-level IDK drops $0.68 \rightarrow 0.26$ and agent accuracy rises $0.23 \rightarrow 0.48$. The communication factor switches from a null lever to a live one: agent accuracy is 0.38 at degree 2 vs. 0.56 on the mesh, where without mimesis the same sweep is flat.

But not the ceiling. System-level accuracy does not improve — 0.72 (no mimesis) vs. 0.66 (mimesis) at baseline — because the system level asks whether *anyone* holds current evidence, and repetition redistributes evidence without creating it. Mimesis pays at the system level only where redistribution is the binding constraint: in the observation-starved cell ($E = 1$) system accuracy rises $0.46 \rightarrow 0.64$ and system IDK falls $0.29 \rightarrow 0.02$. The regime it cannot rescue is fast forgetting ($c = 1.0$: system IDK 0.33 vs. 0.31) — copies crowd out as fast as originals.

4 Mitigation 2: books

Mimesis is peer-to-peer. The second mitigation is a shared artifact: a *book* any agent can read for one action, fed only by deliberate, costly writes — an agent attempts a write with probability p , an attempt consumes its entire step, and a write commits after W attempts, merging the writer’s current grounded knowledge (newest evidence wins). Reads ground answers like observations. We sweep the toy platform (10 seeds per point) in a scarce-observation regime where books have room to matter.

The book’s value is a race between its write rate and the world’s revision rate. The natural unit is the expected write rate $\omega = p/W$ (completed writes per agent per step). The book’s accuracy gain over the no-books floor is large when $\bar{\lambda} \ll \omega$ ($+0.48$ at $\omega = 10/100$ steps, $\bar{\lambda} = 0.001$) and gone when $\bar{\lambda}/\omega \gtrsim 1$: every condition lands on the floor by $\bar{\lambda} = 0.3$ (Fig. 4, left). Availability behaves completely differently: $P(\text{IDK})$ is set by ω ($0.43/0.63/0.83$ for $\omega = 10, 1, 0.2$ per 100 steps, vs. 0.93 without books) and is *flat* in $\bar{\lambda}$ (Fig. 4, right). Staleness absorbs the entire loss: the book keeps everyone answering while the answers quietly expire — a fast world does not silence the library, it turns it into a confident source of expired facts.

ω is the sufficient statistic. Holding p fixed and varying W (constant budget tax) reproduces the ordering, confirming W is a pure freshness dial with diminishing returns once the book turns over faster than the world. Conversely, holding ω fixed while varying p by 4–10 $\times$ moves the curves by at most

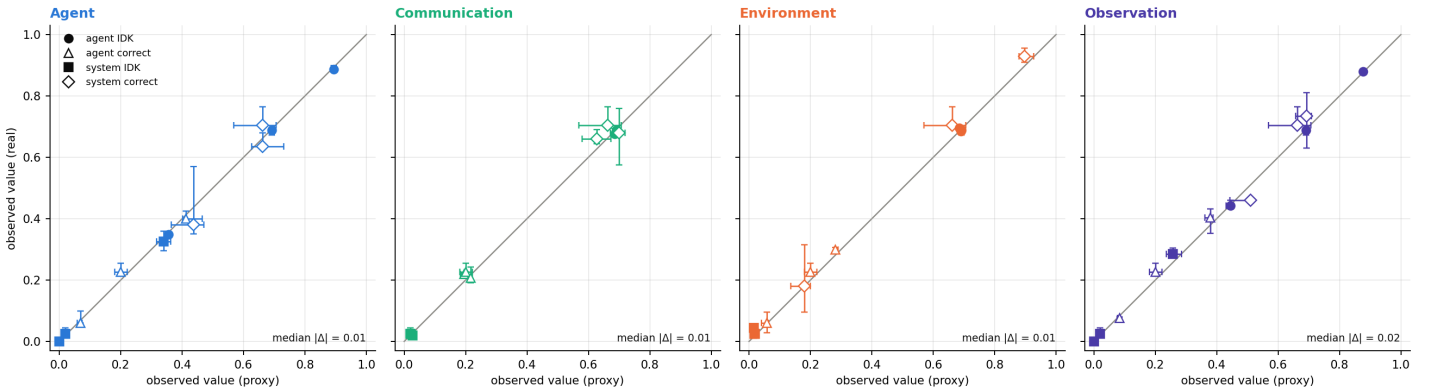


Figure 2: Validation of the proxy: the simulation matches the real system, per factor, with zero fitted parameters. Each panel: that factor’s three settings $\times$ four metrics, plotted as simulation (horizontal, median with 10-seed IQR) against real (vertical, median with seed range). The chain is: measured answering gate $P(\text{response category} \mid \text{context state})$ + verbatim relay + mechanical retrieval. Median $|\Delta| = 0.01$ in every factor; the same holds for the mimesis grid (not shown).

0.06 on either metric, against between- ω gaps up to 0.22. The residual is monotone in p with the sign a budget tax predicts (high- p writers lose more steps to attempts and deposit slightly staler content, ≈ 0.04 – 0.06 accuracy at slow speeds). For design purposes: quote a book policy by its expected writes; the attempt probability is a second-order tax.

control chart on the probe panel see confident staleness coming, given it presents as drift, not shift?

Takeaway and next steps. Communication — secondhand sharing or shared artifacts — reliably buys *availability*, and buys *accuracy* only while the underlying evidence is younger than the world’s revision clock: mimesis is capped by what the population has observed, and a book by how often anyone can afford to write. As the environment speeds up, the failure mode is not silence but confident staleness, which availability metrics cannot see. Next: (1) “dated editions” — books exposing their write time so readers can abstain on stale entries, trading availability back for accuracy; (2) strategy-sensitivity sweeps on the validated platform; (3) the monitoring question: does a

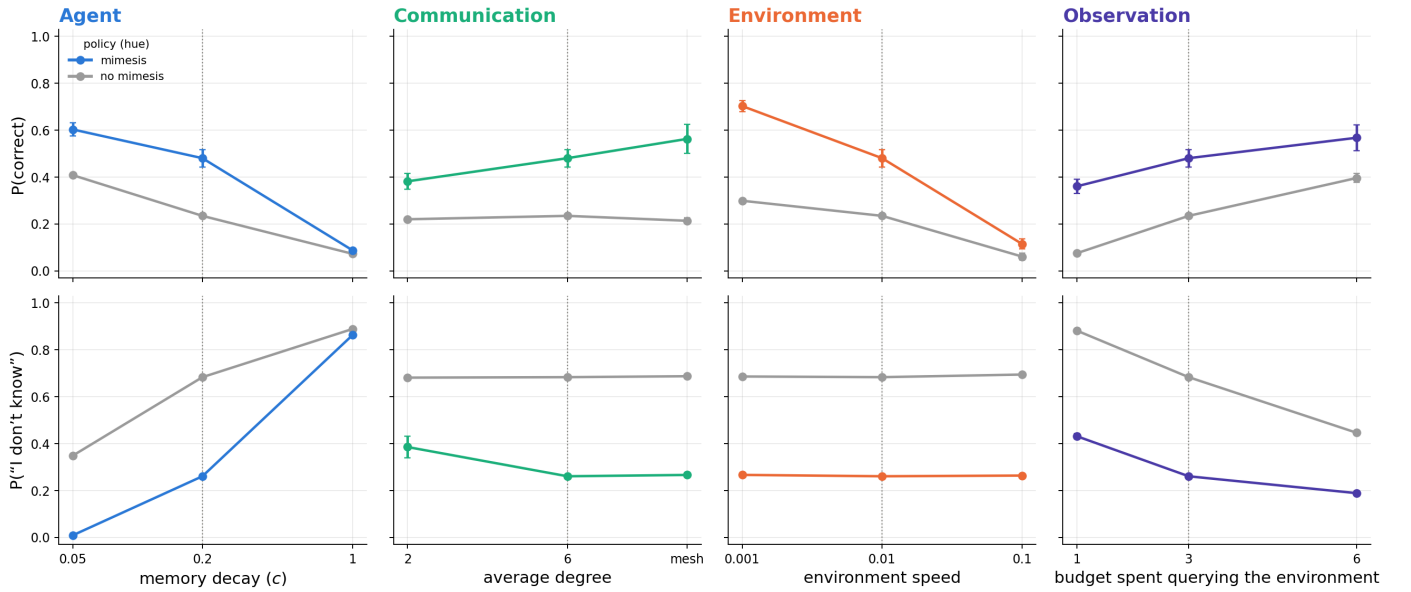


Figure 3: Sharing secondhand information raises the floor, not the ceiling (agent level; same factor grids and protocol as Sec. 2, mimesis vs. no mimesis overlaid). Mimesis roughly halves abstention and doubles accuracy for the average member at baseline and brings the communication graph into play; system-level accuracy (not shown) stays environment-capped: 0.72 vs. 0.66 at baseline.

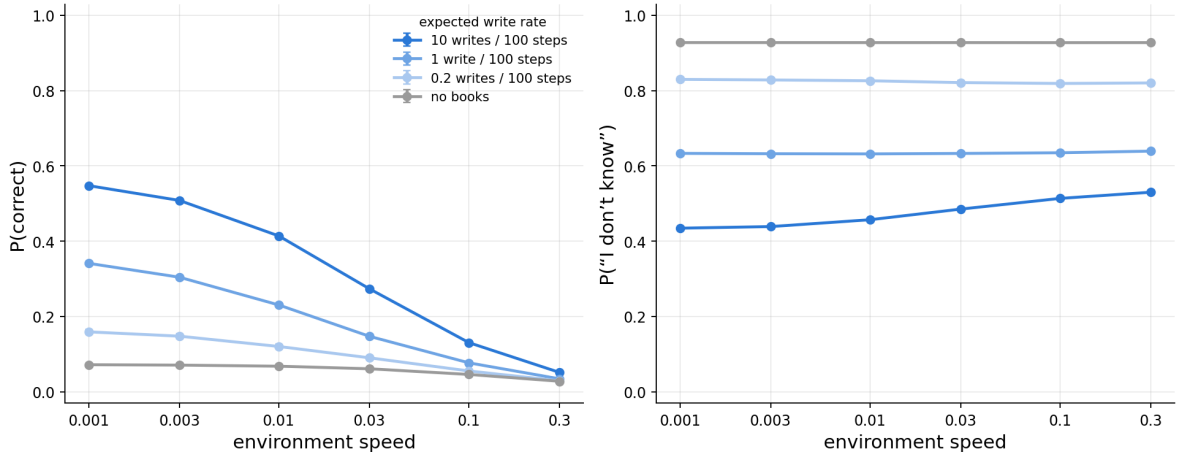


Figure 4: A book buys availability at any speed; its accuracy dies at $\bar{\lambda}/\omega \sim 1$. Toy platform, no mimesis, scarce observation; conditions labeled by expected write rate $\omega = p/W$. Left: accuracy with the book converges to the no-books floor as $\bar{\lambda}$ rises, for every ω . Right: P(IDK) is set by ω and essentially flat in $\bar{\lambda}$.